

Module 54 – Modules

1. What are Modules?

Modules are separate JavaScript files used to organize code into smaller, reusable parts. They allow us to share code between files using export and import.

- **export** → Sends code from a module.
- **import** → Brings exported code into another module.

2. Types of Export

- **Named Export** → Exports specific variables, functions, or classes using their names.
- **Default Export** → Exports one main value from a module and can be imported with any name.

Example Code:

- I. **Using .mjs Extension** → .mjs files are treated as JavaScript modules by default, so they can directly use import and export.

Main.mjs:

```
// import { loginData as newPerson } from './loginPage.mjs';
// import PersonObject from './signUpPage.mjs';
import { newEmployeeId, loginInfo } from './loginPage.mjs';
import signUpInfo from './signUpPage.mjs';
function application(){
    // console.log(newPerson);
    // console.log(PersonObject);
    console.log(newEmployeeId);
    loginInfo();
    signUpInfo();
}
application();
```

LoginPage.mjs:

```
// Named Export
// export let loginData = {
//   uName : "Deveeswar",
//   role : "Developer"
// }
export let newEmployeeId = "UQI12345";
let loginData = {
  uName : "Deveeswar",
  role : "Developer"
};
export function loginInfo(){
  console.log(loginData);
}
```

SignUpPage.mjs

```
// Default Export
let signUpData = {
  uName : "Johny",
  uEmail : "abc@gmail.com",
  age : 45
};
// export default signUpData;
export default function signUpInfo(){
  console.log(signUpData);
}
```

II. **Using .js with package.json** → .js file can be treated as a module by setting "type": "module" in package.json.

Index.js

```
import { products } from "./wishlist.js";  
function indexFile(){  
    console.log(products);  
}  
indexFile();
```

Wishlist.js

```
export let products = [  
    {id : 1, pName : "Mobile"}  
];
```

Package.json

```
{  
  "name": "javascript",  
  "version": "1.0.0",  
  "description": "Javascript Series",  
  "license": "ISC",  
  "author": "Deveeswar",  
  "type": "module",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  }  
}
```

III. **Using .js with index.html** → .js file can be treated as a module in HTML by using type="module" in the <script> tag.

Index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Modules</title>

</head>

<body>

  <h1>Modules in Javascript</h1>

  <!-- type="module" allows import/export in JavaScript -->

  <script type="module" src="products.js"></script>

</body>

</html>
```

Products.js

```
import { products } from "./userData.js";
console.log(products);
```

UserData.js

```
export let products = [
  { id: 1, pName: "Mobile" },
  { id: 2, pName: "Laptop" }
];
```